

Database Design

Business Rules

1. A client can have one or more accounts.
2. An account can belong to one or more clients.
3. A client can have one or more physical addresses.
4. A physical address can belong to one or more clients.
5. An account can have one or more cards.
6. A card can belong to one or more accounts.

Entities & Attributes

Entity: Client

Attributes:

- First name
- Middle name
- Last name
- Date of birth
- Gender
- Nationality
- Identification number
- Phone number
- Email address
- Physical address ID
- Mobile app password

Entity: Client_Address

Attributes:

- Client ID
- Physical address ID

Entity: Physical Address

Attributes:

- Client ID
- Street number
- Street name
- Suburb
- Town
- Province
- Postal code
- Country

Entity: Account_Holder

Attributes:

- Client ID
- Account number

- Card number
- Last access date
- Last deposit date
- Last withdrawal date

Entity: Account

Attributes:

- Account number
- Account type (savings/cheque/credit/business)
- Branch code
- Account status (active/blocked)

Entity: Card

Attributes:

- Card number
- Card pin
- Card type (debit/credit/prepaid)
- Card network/provider (Visa / Mastercard)
- Issuing bank
- Expiry date
- Card status (active/blocked/expired)
- Card verification value (CVV)

Entity Relationships

1. A CLIENT can have one or more accounts; an ACCOUNT can belong to one or more clients. [M:N]

- Entities:
 - Client
 - Account
- Bridge Entity:
 - Account_Holder

2. A CLIENT can have one or more physical addresses; a PHYSICAL ADDRESS can belong to one or more clients. [M:N]

- Entities:
 - Client
 - Physical_Address
- Bridge Entity:
 - Client_Address

3. An ACCOUNT can have one or more cards; a CARD can belong to one or more accounts. [M:N]

- Entities:
 - Account
 - Card
- Bridge Entity:
 - Account_Holder